Technische Universität Clausthal

Optical properties of natural glasses

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Samples


indochinite


moldavite


obsidian

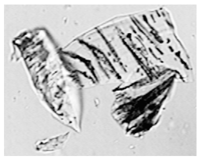

pitchstone

Experimental

The refractive indices have been determined using the immersion method at constant temperature and different wavelengths. Optical dispersion is calculated according to

Abbe $\nu = \frac{n_d - 1}{n_f - n_c}$ and

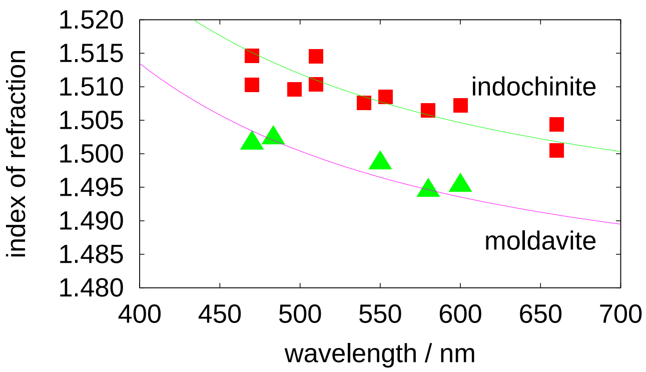
Sellmeier $n(\lambda) = \sqrt{\frac{b_1 \cdot \lambda^2}{\lambda^2 - c_1} + \frac{b_2 \cdot \lambda^2}{\lambda^2 - c_2} + \frac{b_3 \cdot \lambda^2}{\lambda^2 - c_3} + 1}$.

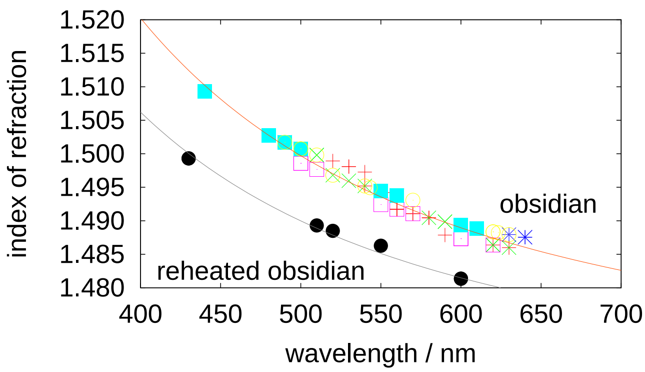

Magnetite crystals inside in obsidian
$\varnothing \approx 1\ \mu$m,
edge $\approx$100 μm

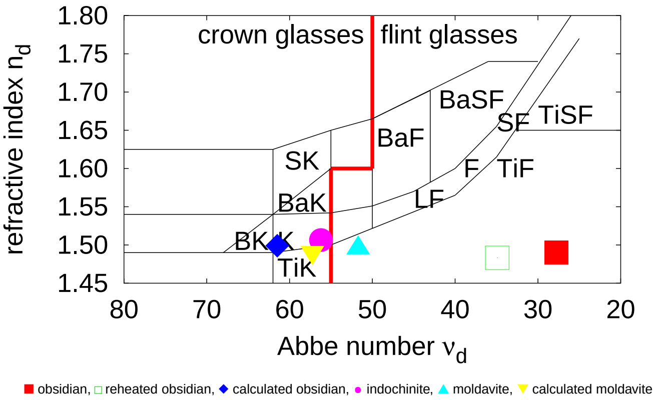
Chemical composition (wt.-%) and optical properties

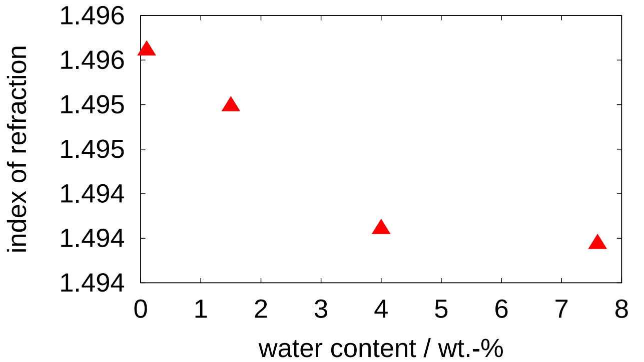
	SiO_2	TiO_2	Al_2O_3	FeO	MnO	MgO	CaO	Na_2O	K_2O	D	$n_{meas.}$	$n^5_{calc.}$	$\nu_{meas.}$	$\nu^5_{calc.}$
indochinite	73.79	0.73	12.48	4.96	0.10	2.26	1.93	1.40	2.43	2.42	1.507	–	56	–
moldavite[1]	78.6	0.31	10.1	1.62	0.09	2.33	2.98	0.42	3.40	2.37	1.495	1.491	52	57
obsidian[2]	73.8	0.13	13.77	1.71	0.06	0.29	1.09	3.97	3.98	2.37	–	–	–	–
obsidian[3]	74.79	0.14	13.78	0.88	0.05	0.13	0.83	3.93	4.96	2.36	1.490	1.499	28	64 (47[6])
reheated obsidian[3]											1.494	–	35	–
remelted pitchstone[4]	72.3	0.09	12.1	0.85	–	0.22	1.13	3.37	2.55	2.34	1.494	–	(17)	–

[1]Bohemia, Czech Rep., [2]Armenia, [3]Ikizdere, Turkey, [4]Meissen, Saxony, 7.6 wt.-% water content, [5]acc. to Appen, [6]acc. to Huggins









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